1. INTRODUCTION

1.1 Project Overview

Electricity is one of the most important resources for the development of a country. Analysing electricity consumption helps governments, electricity boards, and researchers understand usage patterns and make better decisions.

This project, "Plugging into the Future: An Exploration of Electricity Consumption Patterns," analyses electricity consumption data using Tableau Desktop. Interactive dashboards and visualizations are created to compare electricity usage across different states, regions, months, quarters, and years. The project also studies the impact of the COVID-19 lockdown on electricity consumption and presents the results through an interactive Tableau Story.

1.2 Purpose

The purpose of this project is to analyse electricity consumption data and present meaningful insights using Tableau Desktop.

The main objectives are:

- **Analyse electricity consumption across different states and regions.**
- **Compare electricity usage for the years 2019 and 2020.**
- **Identify the highest and lowest electricity-consuming states.**
- **Analyse monthly and quarterly electricity consumption trends.**
- **Study the impact of the COVID-19 lockdown on electricity usage.**
- **Develop interactive dashboards and a Tableau Story.**
- **Support better decision-making through data visualization.**

---

2. IDEATION PHASE

2.1 Problem Statement

Analysing large amounts of electricity consumption data manually is difficult and time-consuming. It is challenging to identify consumption trends, compare different states and regions, and understand changes during events such as the COVID-19 lockdown.

This project provides an interactive Tableau dashboard that helps users easily analyse electricity consumption patterns, compare different regions, and make informed decisions using data visualization.

2.2 Empathy Map Canvas

User

- Government Agencies
- Electricity Boards
- Energy Analysts
- Researchers
- Policymakers

Says

- We need accurate electricity consumption reports.
- We want to compare electricity usage across different states.
- We need interactive dashboards for quick analysis.
- We require reliable data for planning future energy requirements.

Thinks

- Which states consume the most electricity?
- How did electricity consumption change during the COVID-19 lockdown?
- Which region has the highest electricity demand?
- How can electricity resources be managed more efficiently?

Does

- Analyses electricity consumption reports.
- Compares regional and state-wise electricity usage.
- Uses dashboards to monitor electricity trends.
- Makes data-driven decisions for energy planning.

Feels

- Needs accurate and reliable data.
- Wants easy-to-understand visualizations.
- Prefers interactive dashboards over manual reports.
- Seeks faster access to meaningful insights.

2.3 Brainstorming

Several ideas were discussed before selecting the final solution. The objective was to develop an easy-to-use data visualization system capable of presenting electricity consumption patterns effectively.

Ideas Considered

- State-wise electricity consumption analysis.
- Region-wise electricity consumption comparison.
- Monthly electricity consumption trends.
- Quarterly electricity usage analysis.
- Year-wise electricity consumption comparison.
- Top and Bottom electricity-consuming states.
- COVID-19 Lockdown impact analysis.
- Interactive dashboards with dynamic filters.
- Tableau Story for presenting the complete analysis.

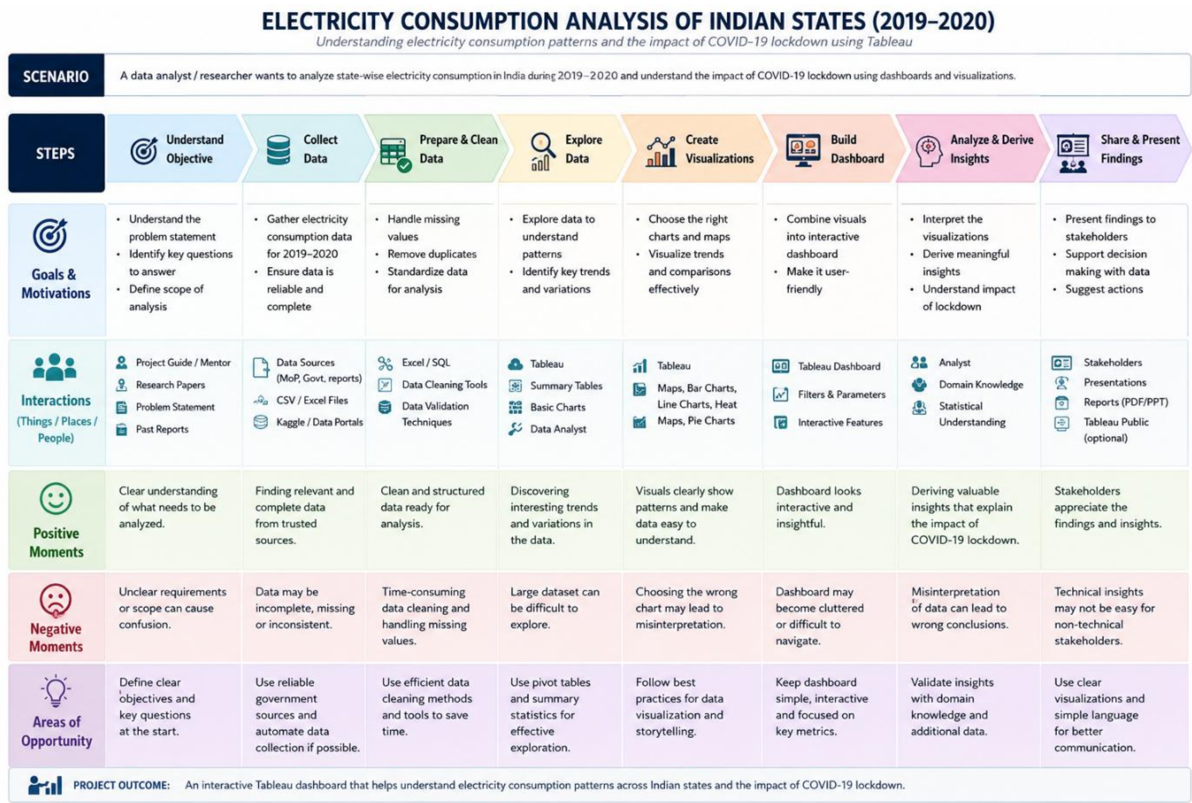
Final Idea

The final solution is **"Plugging into the Future: An Exploration of Electricity Consumption Patterns."** The project uses Tableau Desktop to transform electricity consumption data into interactive dashboards and a Tableau Story. The solution enables users to analyze electricity consumption across different states, regions, months, quarters, and years while providing meaningful insights through interactive visualizations. This approach supports better energy planning, policy-making, and data-driven decision-making.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map describes how users interact with the electricity consumption analysis dashboard from identifying their needs to making informed decisions.



Stage	User Activity	User Experience
Awareness	User wants to analyze electricity consumption data.	Understands the need for an interactive analytics solution.
Exploration	Opens the Tableau dashboard and reviews available visualizations.	Easily navigates through different dashboards.
Interaction	Applies filters such as Year, State, Region, Quarter, Month, Lockdown Period, and Recovery Phase.	Quickly explores electricity consumption patterns.
Analysis	Compares electricity consumption across states, regions, months, and quarters.	Identifies important trends and consumption patterns.

Stage	User Activity	User Experience
Decision Making	Uses dashboard insights for energy planning and policy decisions.	Makes informed, data-driven decisions.
Outcome	Gains meaningful insights from interactive dashboards and Tableau Story.	Improved understanding of electricity consumption patterns.

3.2 Solution Requirements

Functional Requirements

FR No.	Functional Requirement	Description
FR-1	Data Import	Import electricity consumption dataset into Tableau Desktop.
FR-2	Data Preparation	Validate, clean, and preprocess the dataset.
FR-3	Data Visualization	Create interactive charts and visualizations.
FR-4	State-wise Analysis	Analyze electricity consumption across different states.
FR-5	Region-wise Analysis	Compare electricity consumption across regions.
FR-6	Monthly & Quarterly Analysis	Analyze electricity consumption trends over time.
FR-7	Lockdown Analysis	Compare electricity usage before, during, and after COVID-19 lockdown.
FR-8	Dashboard Development	Develop interactive dashboards with filters.
FR-9	Tableau Story	Present dashboards in a logical sequence using Tableau Story.
FR-10	Dashboard Publishing	Publish dashboards on Tableau Public.

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	User-friendly dashboards with interactive filters.
NFR-2	Performance	Fast rendering of visualizations and dashboards.
NFR-3	Reliability	Accurate visualization and consistent dashboard performance.
NFR-4	Security	Secure handling of project data and Tableau workbook.
NFR-5	Scalability	Ability to add new datasets, years, and visualizations without redesigning the dashboard.
NFR-6	Availability	Dashboard accessible through Tableau Public anytime.

3.3 Data Flow Diagram

The Data Flow Diagram illustrates the flow of electricity consumption data from the dataset to the final interactive dashboards.

Process Flow

Electricity Consumption Dataset (CSV)

↓

Data Import

↓

Data Cleaning & Preprocessing

↓

Data Validation & Formatting

↓

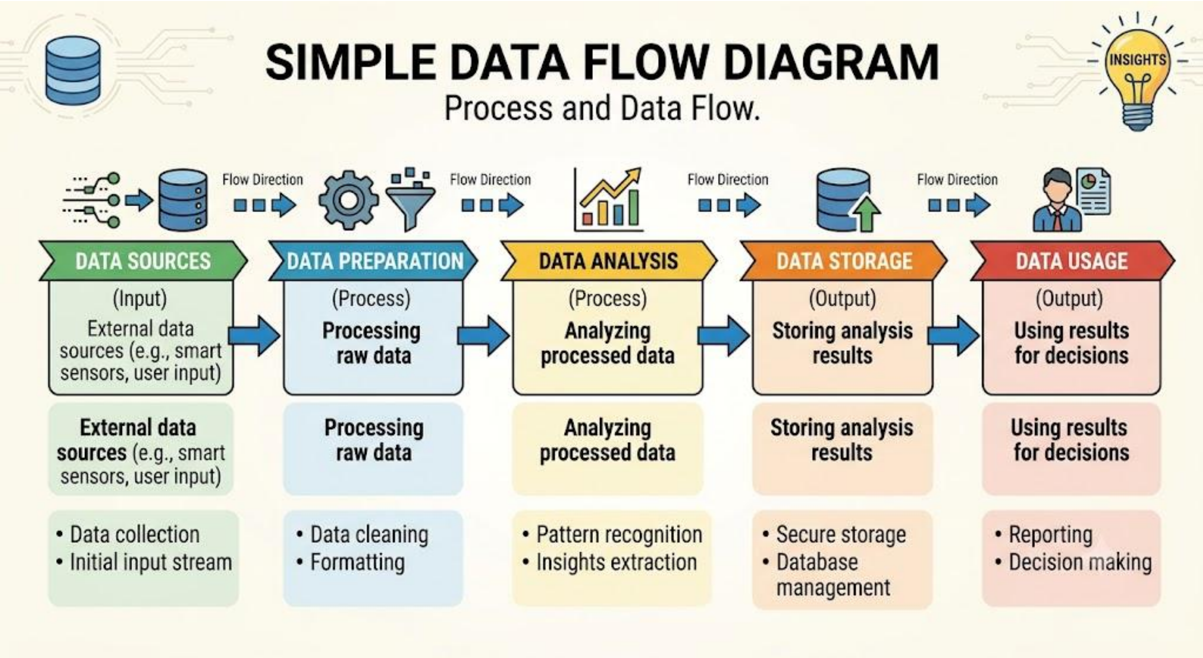
Calculated Fields & Filters

↓

Tableau Visualizations

↓

Interactive Dashboards& Tableau Story



3.4 Technology Stack

Component	**Technology Used**
Programming Language	Not Applicable (Tableau Project)
Data Visualization Tool	Tableau Desktop
Dataset Format	CSV (Comma Separated Values)
Data Preparation	Tableau Desktop
Dashboard Development	Tableau Desktop
Story Development	Tableau Story
Data Publishing	Tableau Public
Operating System	Windows
Dataset	Electricity Consumption Dataset
Visualization Techniques	Filled Map, Bar Chart, Pie Chart, Line Chart
Interactive Features	Filters, Legends, Tooltips, Calculated Fields

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution helps users analyze electricity consumption data in an easy and interactive way. Instead of using manual reports, the project uses Tableau dashboards to visualize electricity consumption across different states, regions, months, quarters, and years. Interactive filters and charts help users identify trends and make better decisions.

Problem Solution Fit

Problem	Proposed Solution
Difficulty in analyzing large electricity consumption data	Interactive Tableau dashboards simplify data analysis.
Difficulty in comparing state-wise and region-wise electricity consumption	Filled maps and charts provide easy comparison.
Difficulty in identifying monthly and quarterly trends	Line charts and bar charts show consumption trends.
Lack of interactive analysis	Filters and calculated fields allow dynamic analysis.
Difficulty in understanding the impact of COVID-19	Lockdown and Recovery Phase analysis shows changes in electricity consumption.

4.2 Proposed Solution

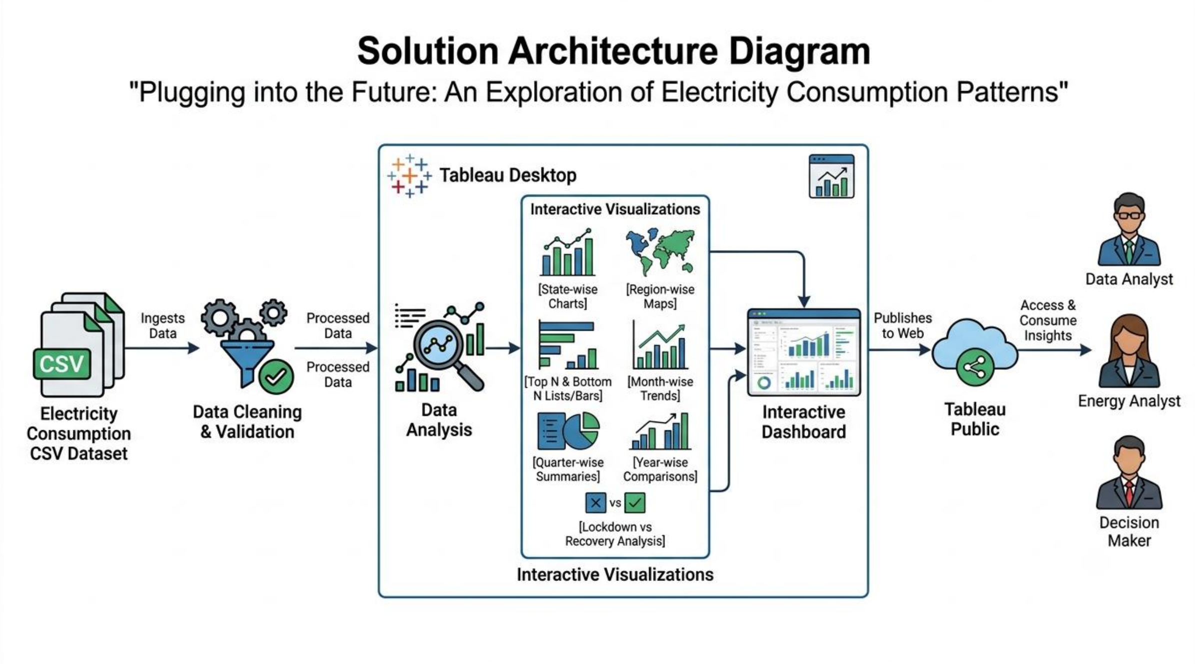
The proposed solution is an interactive electricity consumption analysis system developed using Tableau Desktop. The system imports electricity consumption data, performs preprocessing, creates calculated fields, and generates meaningful visualizations. Multiple dashboards are developed to analyze state-wise, region-wise, quarterly, monthly, and yearly electricity consumption patterns.

Interactive filters such as **Year**, **State**, **Region**, **Month**, **Quarter**, **Lockdown Period**, and **Recovery Phase** enable users to dynamically explore the dataset. A Tableau Story combines all dashboards into a structured presentation, allowing users to understand the complete analysis step by step.

The final workbook is published on Tableau Public, making it accessible online for users to interact with the dashboards and story.

4.3 Solution Architecture

The solution architecture illustrates the complete workflow of the project from data collection to publishing the interactive dashboards.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project planning plays a crucial role in ensuring the successful completion of the Electricity Consumption Analysis project. The project was planned using Agile methodology by dividing the work into multiple sprints. Each sprint focused on a specific set of tasks, including data collection, preprocessing, visualization, dashboard development, story creation, testing, and publishing. Story points were assigned based on the complexity and effort required for each task.

Agile Sprint Planning

Sprint

A Sprint is a fixed period during which a team completes a set of predefined tasks.

Epic

An Epic is a large feature that is divided into multiple smaller User Stories.

User Story

A User Story represents a specific functionality or task that contributes to completing an Epic.

Story Point Scale

Story Point	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult

Sprint Planning

Sprint 1

Epic 1 – Data Collection

User Story	Story Points
Import Electricity Consumption Dataset	2
Validate Dataset	2

Epic 2 – Data Preparation

User Story	Story Points
Clean Missing Values	3
Create Calculated Fields	3
Organize and Format Data	2

Total Story Points (Sprint 1) = 12

Sprint 2

Epic 3 – Data Visualization

User Story	Story Points
State-wise Analysis Dashboard	3
Region-wise Analysis Dashboard	3

User Story	Story Points
Month-wise & Quarter-wise Analysis	5
Lockdown & Recovery Analysis	3

Epic 4 – Dashboard Development

User Story	Story Points
Develop Interactive Dashboards	3

Epic 5 – Story Development

User Story	Story Points
Create Tableau Story & Publish	3

Total Story Points (Sprint 2) = 20

Velocity Calculation

Total Story Points = 12 + 20 = 32

Number of Sprints = 2

Velocity = Total Story Points / Number of Sprints

Velocity = 32 / 2 = 16 Story Points per Sprint

Team Velocity = 16 Story Points per Sprint

Sprint Backlog

Sprint	Functional Requirement	User Story No.	Story Points	Priority
Sprint 1	Data Collection	USN-1	2	High
Sprint 1	Data Collection	USN-2	2	High
Sprint 1	Data Preparation	USN-3	3	High
Sprint 1	Data Preparation	USN-4	3	Medium
Sprint 1	Data Preparation	USN-5	2	Medium

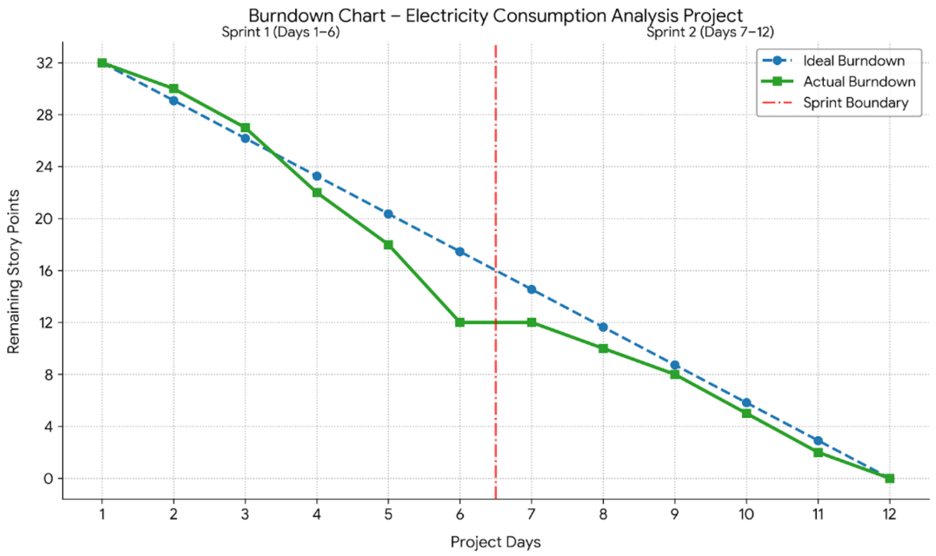
Sprint	Functional Requirement	User Story No.	Story Points	Priority
Sprint 2	Data Visualization	USN-6	3	High
Sprint 2	Data Visualization	USN-7	3	High
Sprint 2	Data Visualization	USN-8	5	High
Sprint 2	Data Visualization	USN-9	3	High
Sprint 2	Dashboard Development	USN-10	3	High
Sprint 2	Story Development	USN-11	3	Medium

Project Tracker

Sprint	Total Story Points	Duration	Story Points Completed
Sprint 1	12	6 Days	12
Sprint 2	20	6 Days	20

Burndown Chart

A burndown chart was created to monitor project progress by comparing the planned story points with the completed story points during each sprint. The chart demonstrates that all planned tasks were completed within the estimated sprint duration.



6. FUNCTIONAL AND PERFORMANCE TESTING

Testing was performed to verify that all project functionalities worked correctly and that the Tableau dashboards provided accurate and interactive visualizations. Both functional testing and performance testing were carried out to ensure the reliability, usability, and efficiency of the solution.

6.1 Functional Testing

Functional Testing Table

Test Case ID	Functionality Tested	Expected Result	Actual Result	Status
FT-01	Import Electricity Consumption Dataset	Dataset should load successfully into Tableau	Successfully imported	Pass
FT-02	Data Preprocessing	Dataset should be cleaned and validated	Successfully completed	Pass
FT-03	Calculated Fields	Calculated fields should generate correct values	Generated correctly	Pass
FT-04	Interactive Filters	Filters should update all visualizations dynamically	Working correctly	Pass
FT-05	Filled Maps	Maps should display state-wise electricity consumption accurately	Displayed correctly	Pass
FT-06	Charts & Graphs	Charts should represent accurate electricity consumption data	Working correctly	Pass
FT-07	Interactive Dashboards	Dashboards should display all visualizations without errors	Working correctly	Pass
FT-08	Tableau Story	Story should navigate between dashboards correctly	Working correctly	Pass
FT-09	Tableau Public Publishing	Dashboard should be accessible online	Successfully published	Pass

---

6.2 Performance Testing

The developed Tableau dashboards were evaluated based on different performance parameters to ensure efficient data visualization and user interaction.

Parameter	**Observation**	**Status**
Dashboard Loading Time	Dashboards loaded successfully without noticeable delay	Pass
Filter Response Time	Interactive filters updated visualizations quickly	Pass
Data Accuracy	All charts displayed accurate electricity consumption values	Pass
Dashboard Navigation	Smooth navigation between dashboards and story points	Pass
Tableau Public Access	Dashboard accessible through Tableau Public	Pass

---

6.3 Usability Testing

The dashboard was designed with a simple and user-friendly interface. Interactive filters, tooltips, legends, and clear chart titles allow users to explore electricity consumption data without technical expertise.

Result: Successfully Passed

6.4 Scalability Testing

The solution is scalable and can accommodate additional electricity consumption data, new years, new states, and more visualizations without changing the overall dashboard architecture.

Result: Successfully Passed

6.5 Reliability Testing

The dashboard consistently displayed accurate results throughout testing. All visualizations responded correctly to user interactions and filters.

Result: Successfully Passed

6.6 Availability Testing

The project was successfully published on Tableau Public, making it accessible online anytime through the published workbook link.

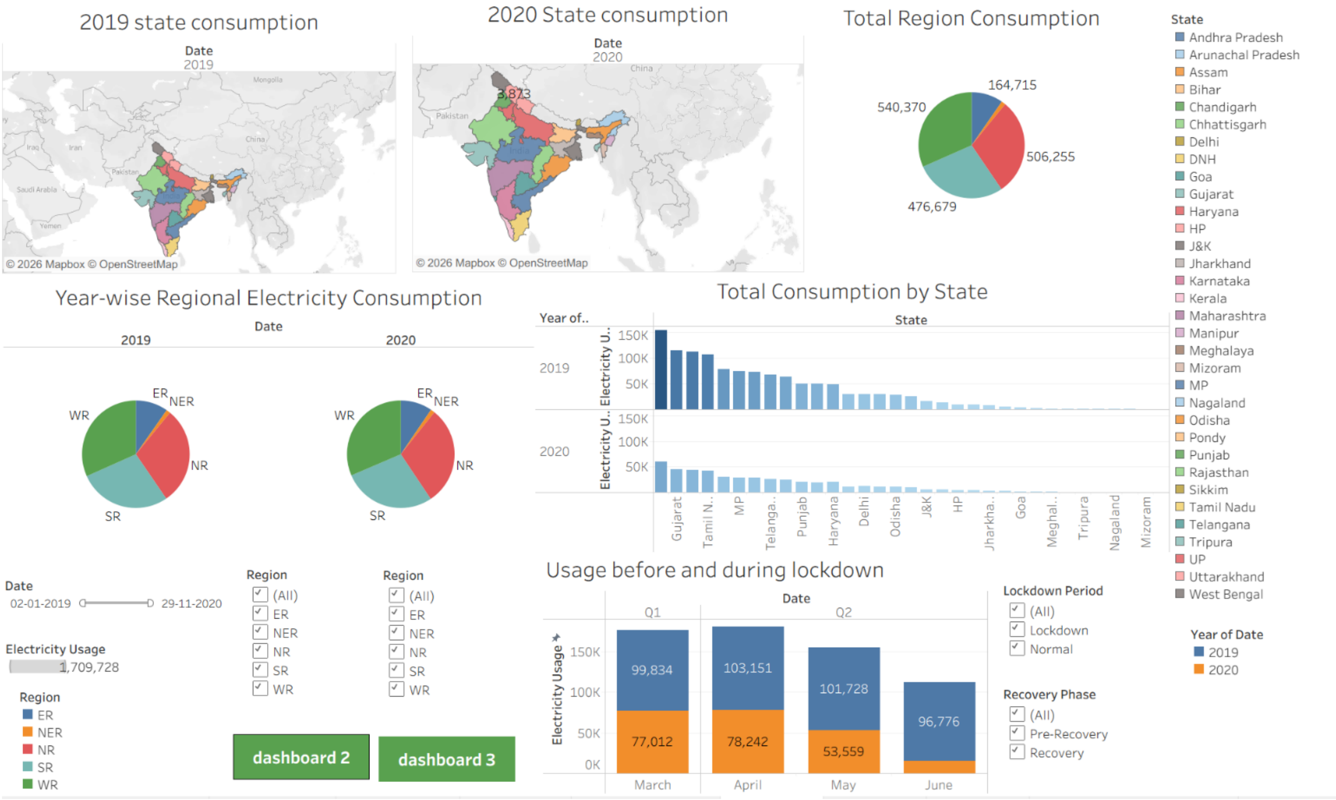
Result: Successfully Passed

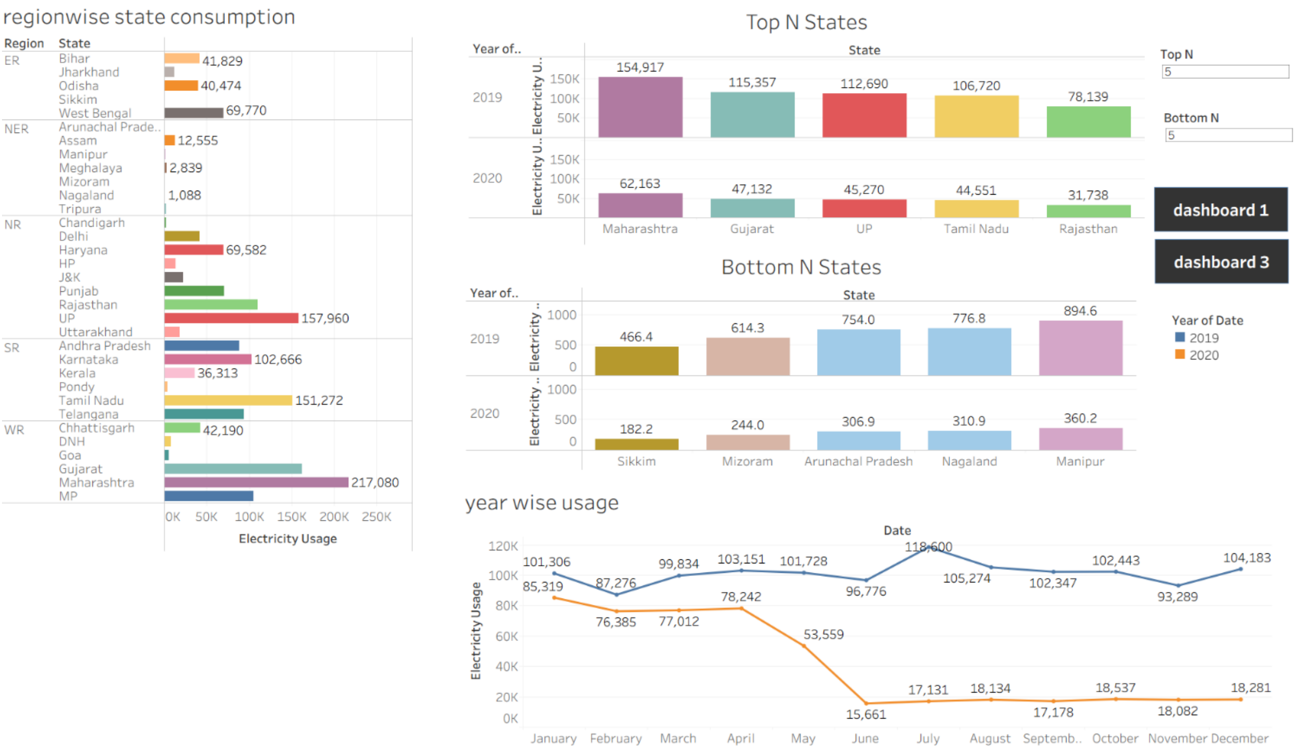
6.7 Performance Summary

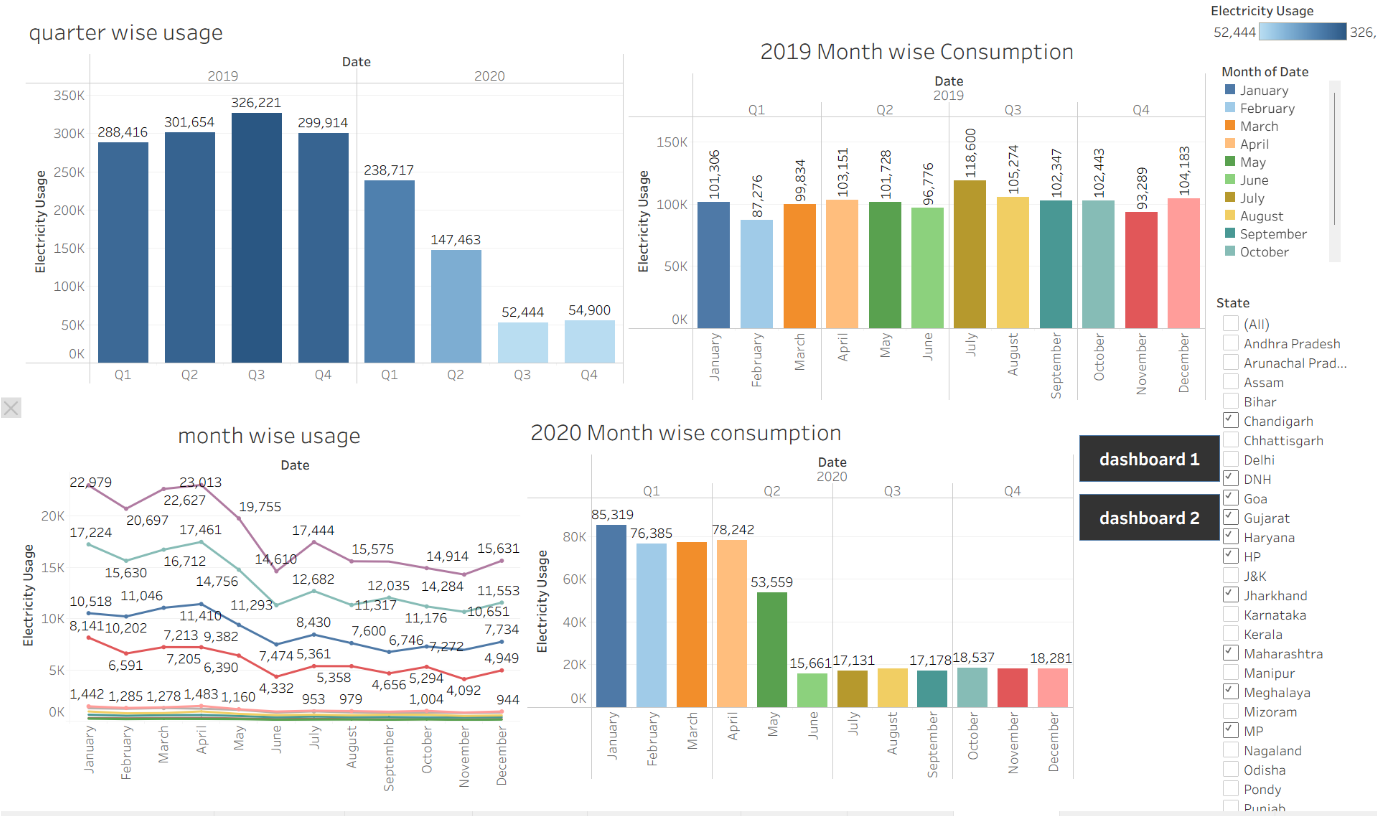
Characteristic	Description	Technology Used
Performance	Dashboards load quickly and provide smooth user interaction.	Tableau Desktop, Tableau Public
Usability	Interactive filters and dashboards make data analysis simple and user-friendly.	Tableau Desktop
Reliability	Visualizations accurately represent the electricity consumption dataset.	Tableau Calculated Fields & Visualizations
Scalability	New datasets, years, and visualizations can be added without changing the dashboard structure.	Tableau Dashboard
Availability	Published dashboards are accessible online through Tableau Public.	Tableau Public
Maintainability	Dashboards can be easily updated when new electricity consumption data is available.	Tableau Desktop

7. RESULTS

The Electricity Consumption Analysis project successfully transformed raw electricity consumption data into meaningful insights using Tableau Desktop. Interactive visualizations, dashboards, and a Tableau Story were developed to analyze electricity consumption across different states, regions, months, quarters, and years. The results demonstrate how data visualization supports better understanding and decision-making in energy management.


2019 state consumption
Date
2019
© 2026 Mapbox © OpenStreetMap
2020 State consumption
Date
2020
3,873
© 2026 Mapbox © OpenStreetMap
Total Region Consumption
164,715
540,370
506,255
476,679
State
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Chandigarh
Chhattisgarh
Delhi
DNH
Goa
Gujarat
Haryana
HP
J&K
Jharkhand
Karnataka
Kerala
Maharashtra
Manipur
Meghalaya
Mizoram
MP
Nagaland
Odisha
Pondy
Punjab
Rajasthan
Sikkim
Tamil Nadu
Telangana
Tripura
UP
Uttarakhand
West Bengal
Year-wise Regional Electricity Consumption
Date
2019
2020
ER
NER
WR
NR
SR
Total Consumption by State
Year of..
State
2019
2020
Electricity U..
150K
100K
50K
Gujarat
Tamil N..
MP
Telanga..
Punjab
Haryana
Delhi
Odisha
J&K
HP
Jharkha..
Goa
Meghal..
Tripura
Nagaland
Mizoram
Date
02-01-2019
29-11-2020
Electricity Usage
1,709,728
Region
ER
NER
NR
SR
WR
Region
(All)
ER
NER
NR
SR
WR
dashboard 2
dashboard 3
Usage before and during lockdown
Date
Q1
Q2
Electricity Usage
150K
100K
50K
0K
99,834
77,012
103,151
78,242
101,728
53,559
96,776
March
April
May
June
Lockdown Period
(All)
Lockdown
Normal
Recovery Phase
(All)
Pre-Recovery
Recovery
Year of Date
2019
2020


regionwise state consumption
Region
State
ER
Bihar
Jharkhand
Odisha
Sikkim
West Bengal
41,829
40,474
69,770
NER
Arunachal Prade..
Assam
Manipur
Meghalaya
Mizoram
Nagaland
Tripura
12,555
2,839
1,088
NR
Chandigarh
Delhi
Haryana
HP
J&K
Punjab
Rajasthan
UP
Uttarakhand
69,582
157,960
SR
Andhra Pradesh
Karnataka
Kerala
Pondy
Tamil Nadu
Telangana
102,666
36,313
151,272
WR
Chhattisgarh
DNH
Goa
Gujarat
Maharashtra
MP
42,190
217,080
0K
50K
100K
150K
200K
250K
Electricity Usage
Top N States
Year of..
State
2019
2020
Electricity U..
154,917
115,357
112,690
106,720
78,139
62,163
47,132
45,270
44,551
31,738
Maharashtra
Gujarat
UP
Tamil Nadu
Rajasthan
Top N
5
Bottom N
5
dashboard 1
dashboard 3
Bottom N States
Year of..
State
2019
2020
Electricity
1000
500
0
466.4
614.3
754.0
776.8
894.6
182.2
244.0
306.9
310.9
360.2
Sikkim
Mizoram
Arunachal Pradesh
Nagaland
Manipur
Year of Date
2019
2020
year wise usage
Date
Electricity Usage
120K
100K
80K
60K
40K
20K
0K
101,306
85,319
87,276
99,834
76,385
77,012
103,151
78,242
101,728
53,559
96,776
118,600
15,661
105,274
17,131
18,134
102,347
17,178
102,443
18,537
93,289
18,082
104,183
18,281
January
February
March
April
May
June
July
August
Septemb..
October
November
December



8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Easy to analyze electricity consumption data.
- Provides interactive and user-friendly dashboards.
- Supports state-wise, region-wise, monthly, and yearly analysis.
- Interactive filters improve data exploration.
- Helps users make better decisions using visual insights.
- Easily scalable by adding new data.

8.2 Disadvantages

- Depends on the quality of the available dataset.
- Does not support real-time data analysis.
- Requires an internet connection to access Tableau Public dashboards.
- Future consumption prediction is not included.

9. CONCLUSION

The project **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** was successfully developed using Tableau Desktop. Interactive dashboards and a Tableau Story were created to analyze electricity consumption across different states, regions, months, and years. The project provides clear insights into electricity consumption patterns and helps users make informed decisions through data visualization.

---

10. FUTURE SCOPE

The project can be improved in the future by:

- Adding real-time electricity consumption data.
- Using machine learning for electricity demand prediction.
- Including renewable energy data.
- Developing mobile-friendly dashboards.
- Integrating live databases for automatic updates.

11. APPENDIX

Tableau Public Link

https://public.tableau.com/app/profile/jagarlamudi.sree.harshini/viz/Electricity_Consumption_Analysis_17837570329990/StoryonelectricityConsumption

GitHub Repository Link

kandlaguntavarun4/Electricity_Consumption_Analysis

Project Demo Link

https://drive.google.com/file/d/1rHdgAFOVnPvPMAKtxHsKTEyJuarsTohz/view?usp=drivesdk